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PROFESSIONAL SUMMARY

Dynamic and detail-oriented **Design Engineer** with 5 years of experience in mechanical design, CAD modeling, product development, and quality assurance. Experienced in valve assemblies, CNC machinery, renewable energy components, and structural analysis. Skilled in **FreeCAD, AutoCAD, SolidWorks, STAAD.Pro, Autodesk Fusion 360, ANSYS, Creo, NX CAD**, and other advanced engineering software, with strong capabilities in design validation, documentation, and cross-functional collaboration.

CORE SKILLS

- 3D Modeling & 2D Drafting (SolidWorks, AutoCAD, Fusion 360)
 - Mechanical & Sheet Metal Design
 - GD&T & Tolerance Stack-Up Analysis
 - Product Development & Prototype Testing
 - BOM Preparation & Documentation
 - Structural & Thermal Analysis (STAAD.Pro, FEA basics)
 - CNC & Industrial Equipment Design
 - Project Coordination & Technical Training
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PROFESSIONAL EXPERIENCE

Design Engineer

Microbeam Laser System | Surat | Jul 2024 – Present

- Designing mechanical parts and assemblies for laser systems.
- Preparing detailed 3D models, fabrication drawings, and BOMs.
- Supporting manufacturing teams for design feasibility and accuracy.

Application Engineer

Sopan Institute of Engineering & Design (SIED) | Navsari | Mar 2022 – Apr 2024

- Provided CAD training and technical workshops for students.
- Guided industrial design projects and ensured documentation accuracy.
- Assisted in industry collaboration and practical project implementations.

QC Inspector

ArcelorMittal Nippon Steel India (AMNS) | Hajira, Surat | Mar 2021 – Mar 2022

- Performed quality inspections and dimensional checks on steel products.
 - Prepared inspection reports and coordinated defect resolution with production teams.
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INTERNSHIPS

Intern – Techrest Design Pvt. Ltd. | Vadodara | Mar 2019 – Apr 2019

- Performed 2D drafting and 3D modeling using SolidWorks.
- Conducted assembly validation and design feasibility checks.
- Supported CFD analysis in ANSYS and basic structural analysis in SolidWorks.
- Assisted in topology test, drop test, and design study tasks.

Intern – Sahakari Khand Udyog Mandal Ltd. | Gandevi | Dec 2019 – Dec 2019

- Studied machineries and equipment used in mill section operations.
- Understood boiler and turbine section functioning and layout.
- Gained insights into complete sugar production process.

PROJECTS

1. O2 Valve Assembly – Tidan Forging Pvt. Ltd.

Designed oxygen valve components ensuring safe, controlled flow for medical and industrial systems.

2. N2 Valve Assembly – Tidan Forging Pvt. Ltd.

Engineered nitrogen valve with high-strength body and precise flow regulation for storage/distribution applications.

3. CO2 Valve Assembly – Tidan Forging Pvt. Ltd.

Developed CO₂ valve with gas-compatible sealing and safety features for controlled flow and blockage prevention.

4. Vertical Axis Wind Mill – Paixtech

Collaborated in multidisciplinary team to design and assemble wind turbine prototype maximizing energy capture.

5. Frame Design (Solar Module) – Waree Energies Ltd.

Modeled and refined solar module frames using CAD tools, enhancing durability and manufacturing alignment.

6. Structural Modeling & Analysis – Lifequbes Shelter Pvt. Ltd.

Performed structural analysis of shade with rafters and purlins using STAAD.Pro for safety and stability.

7. Airless Tyre Design – SVNIT Student Project

Conceptualized and modeled non-pneumatic tire system; analyzed deformation under static and dynamic loads.

8. CNC Router Design – Renubhumi

Designed compact CNC router with optimized frame rigidity and precise linear motion components.

9. Scissor Table – RB Industries

Designed scissor lift table mechanism with optimized linkage geometry for smooth vertical lifting and load transmission.

10. Sambrani Cup Mold Design – Individual

Created 3D CAD models and 2D drawings for mold components ensuring manufacturability and accurate fit.

11. Piping Design – Outsource Project

Prepared detailed piping layout drawings in AutoCAD, incorporating valves, fittings, supports, and flow annotations.

12. Heat Sink Design – Paixtech

Designed aluminum heat sink using thermal analysis to optimize fin geometry for heat dissipation.

13. Textile Loom Frame & Part Design – Nileshe & Paresh Industries

Modeled structural frames and individual components of textile loom machines ensuring precise fit and manufacturability.

EDUCATION**Bachelor of Engineering – Mechanical Engineering**

Gujarat Technological University — 2020

CERTIFICATIONS

- CSWA – SolidWorks CAD Design Associate (23/03/2022)
 - SolidWorks Additive Manufacturing Associate (26/03/2022)
 - SolidWorks Sheet Metal Professional (28/04/2022)
 - SolidWorks Sustainability Associate (25/02/2023)
 - CSWP – SolidWorks CAD Design Professional (22/03/2023)
 - SolidWorks Simulation Associate (29/08/2023)
 - Autodesk Certified User (ACU)
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PERSONAL DETAILS

- **Location:** Gandevi, Navsari, Gujarat
- **Languages:** English, Hindi, Gujarati
- **Notice Period:** Immediate